Link adjusted von Mises

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Parametrisation

The (circular) von Mises distribution has density

$$p(y_i) = \frac{\exp(s_i \kappa \cos(x - \mu))}{2\pi I_0(s_i \kappa)}$$

where $s_i > 0$ is a fixed scaling, the location is μ and the precision-parameter is κ.

Link-function

The linkfunction is given as

$$\mu = 2\pi \left(\frac{1}{1 + \exp(-\eta)} - 1/2 \right)$$

where η is the linear predictor.

Hyperparameters

The hyperparameter is $\kappa = \exp(\theta)$ and prior are given on θ

Specification

- `family="vm"`
- Required arguments: y (and optional s through option `scale`)

Hyperparameter spesification and default values

doc `Link adjusted von Mises circular distribution`

hyper

theta

hyperid `109301`
name `precision`
short.name `prec`
output.name.intern `prec_intern for lavm`
output.name `precision parameter for lavm`
initial `2`
fixed `FALSE`
prior `loggamma`
param `1 0.01`
to.theta `function(x) log(x)`
from.theta `function(x) exp(x)`

experimental `TRUE`

survival `FALSE`

discrete `FALSE`

link `default circular tan identity`

pdf `lavm`

Example

In the following example we estimate the parameters in a simulated example.

```
library(circular)

lfun <- function(x) 2 * pi * (1/(1+exp(-x)) - 0.5)
lfuninv <- function(x) log((pi + x)/(pi - x))

n <- 300
x <- rnorm(n, sd = 0.2)
x <- x - mean(x)
mu <- lfun(0.1 + x)
kappa <- 10
y <- numeric(n)
for(i in 1:n) {
    y[i] <- rvonmises(1, circular(mu[i]), kappa)
}
r <- inla(y ~ 1 + x,
          data = data.frame(y, x),
          family = "vm",
          control.family = list(hyper = list(
                                    prec = list(initial = log(kappa),
                                                fixed = TRUE))),
          verbose = TRUE)
summary(r)
```

Notes

This distribution is experimental, and changes will occur.